NitroTron3 — User Manual

DIY digital bass pedal — Daisy Seed + Hothouse

A project of Nitro Mahalia. This pedal packages several of their signature bass-through-synth sounds into a single bass-specific unit and more: three distinct modes, each doing something not easily found in off-the-shelf pedals. Built on the Electro-Smith Daisy Seed and Cleveland Music Co. Hothouse DSP kit.

Modes at a glance

BORDUN (Mode A). A harmonic companion to the bass, modelled after a specific usage of the Moog MoogerFooger FreqBox (MF-107) — its envelope-gated oscillator mixed in alongside the dry signal, kept clean of sync and FM modulation. An internally generated oscillator, gated and shaped by the bass's own envelope, lays subtle or assertive accompanying harmonics over the input — pure intervals, fifths, octaves, drone-like wash. Placed **before** overdrive in the chain it stacks musically into a saturated sound; on its own it sits as a parallel voice along the played notes. Tracking modes lock the harmony to the played pitch; fixed mode anchors a drone against which the bass moves.

SPRAWL (Mode B). Granular-delay-based texture and soundscape engine. A rolling buffer feeds a grain scheduler whose voices can pitch-shift non-linearly, drift, scatter, and stutter — built to fill the void around the bass, intended for improvisation and experimental performance. Functionally a multi-mode effect on its own: all colouring and texturing stages (decimator/fold, event-driven glitch, ringmod, frequency shifter) are reachable in a non-delay path too (K2 at noon). High feedback with the tanh saturator pushes the loop into harmonic cloud blooms; the Bode SSB shifter inside the feedback loop cascades each pass and rapidly grows beyond pitched material.

SCHISM (Mode C). Dynamic bass filter and digital distortion unit, with a fat pitch-tracked synth voice as a third drive option. The filter can self-oscillate in a controlled manner — singing-into-screaming textures that play well into a downstream overdrive or fuzz. The bit-XOR drive enriches overtones cleanly and lights up especially well placed **before** overdrive/fuzz. The phaser sub-mode is provisional and likely to be replaced with a different effect in a future release.

Presets. A global preset system recalls mode + full parameter state in one footswitch press. One global edit buffer, 3 banks × 8 slots = 24 reachable presets. Each slot carries its own mode, so cycling presets can swap mode mid-set. See the *Footswitches and Presets* section below.

Hardware overview

- 6 knobs (K1–K6, left-to-right, top row then bottom)
- 3 three-position toggle switches (SW1–SW3)
- 2 footswitches (FS1 = preset / FS2 = bypass; both held together for bank-cycle / bootloader)
- 2 indicator LEDs

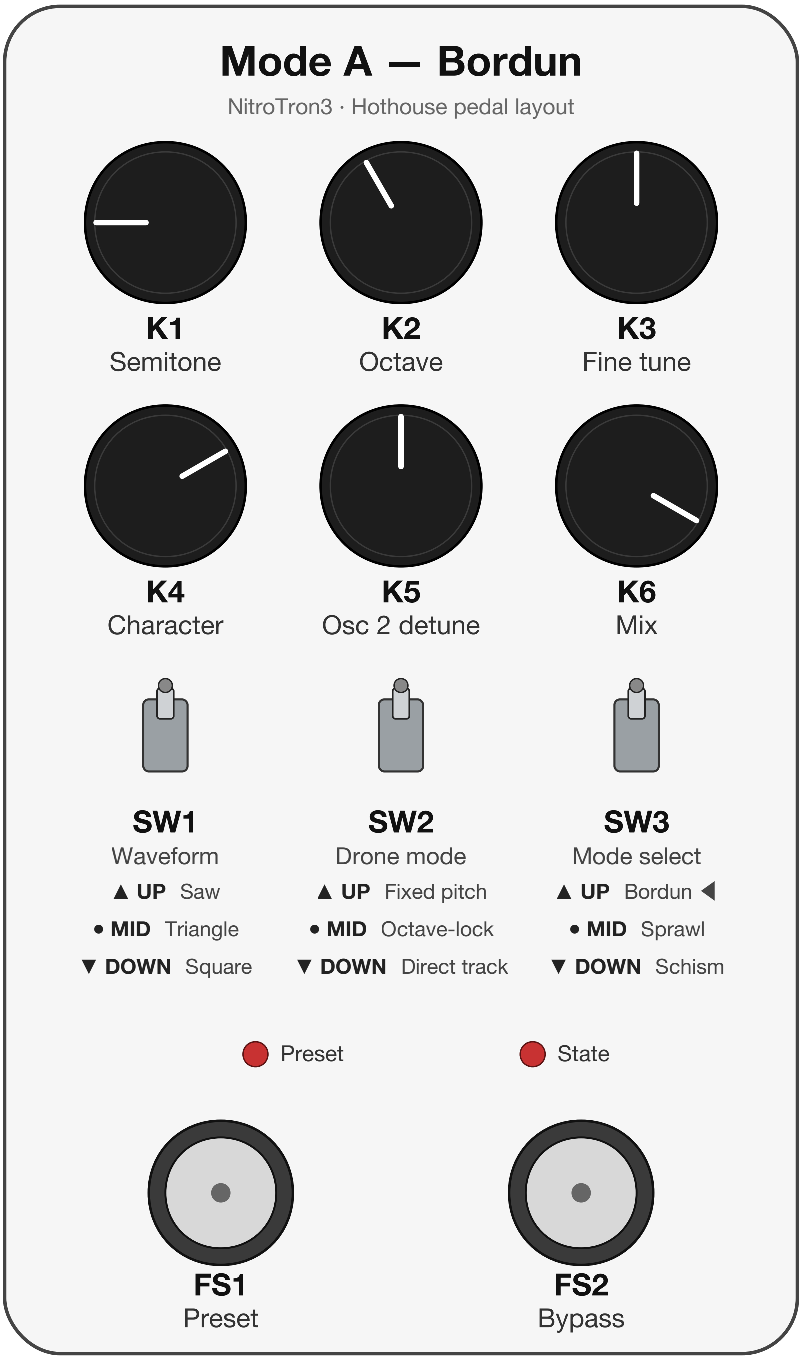
SW3 selects the mode:

- **UP** — BORDUN (Mode A)
- **MIDDLE** — SPRAWL (Mode B)
- **DOWN** — SCHISM (Mode C)

The mode determines what every other knob and switch does. Each mode is documented in its own section below.

BORDUN (Mode A)

A two-oscillator voice (waveform via SW1) tracks the bass input, gated by an envelope follower so the drone only sounds while you play. A Moog ladder filter shapes the tone; in triangle mode K4 crosses over into wavefolding past noon. SW2 picks the pitch source: fixed, octave-locked tracking, or direct tracking.

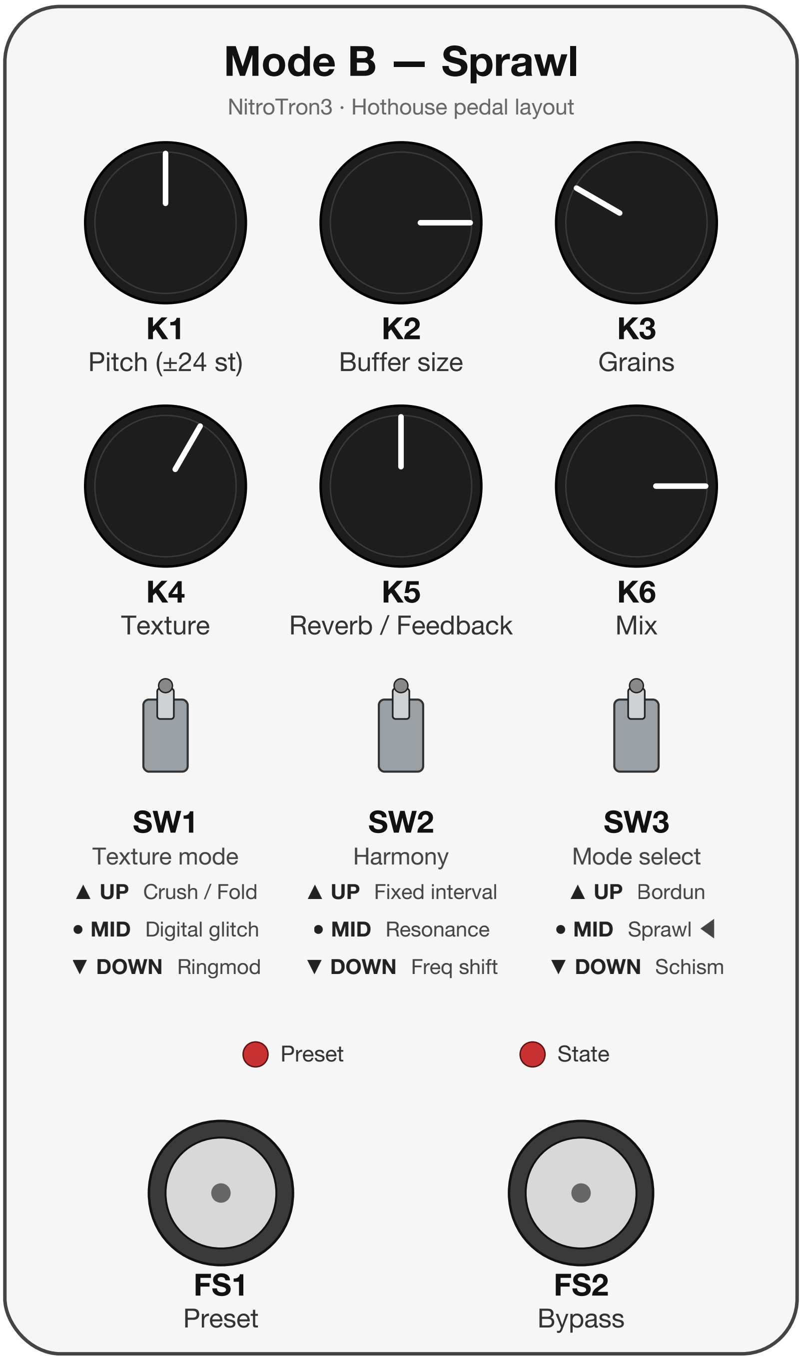


Controls

CONTROL	DESCRIPTION	NOTES
KNOB 1	Semitone / Interval	±12 semitone offset, centered with deadzone. **Fixed**: center = A, also sets the wrap point for tracking. **Track**: center = tracked note, wraps at the note set in fixed mode
KNOB 2	Octave	7 positions (C-1 – C5). **Octave-locked**: sets target octave. **Direct**: ±3 octave offset from tracked pitch
KNOB 3	Fine tune	±50 cents continuous (osc 1 only — creates beating with osc 2)
KNOB 4	Tone / Wavefold	SAW/SQ: full ladder cutoff (80 Hz – 8 kHz). TRI: CCW → noon = cutoff, noon → CW = wavefolding (filter stays fully open)
KNOB 5	Osc 2 detune	Center = off (deadzone). Outside center = ±1–12 semitone steps. Not affected by fine tune
KNOB 6	Mix	0 = full dry, 1 = full wet (oscillator)
SWITCH 1	Waveform	**UP** — Saw • **MIDDLE** — Triangle • **DOWN** — Square
SWITCH 2	Drone mode	**UP** — Fixed pitch (K1 sets note, K2 sets octave) • **MIDDLE** — Octave-locked tracking (pitch class follows bass in K2's octave, K1 adds interval) • **DOWN** — Direct tracking (osc follows exact bass pitch, K1/K2 are relative offsets ±12 semi / ±3 oct)

SPRAWL (Mode B)

8-second SDRAM ring buffer feeding 8 grain voices, with a choice of texture shaper (SW1) and harmony source (SW2). K5 is bipolar — CCW routes the wet bus through a Clouds reverb, CW drives a tanh-saturated feedback loop with build-up and on-play duckers that keep the loop musical. K2 is bipolar around noon: noon bypasses the grain engine and routes the dry through the texture shaper directly (K3 becomes a micro-stutter control), and off noon the sign sets grain playback direction (CW forward, CCW backward). SW2 DOWN replaces grain pitch-shifting with a Bode SSB frequency shifter living inside the feedback loop.



Controls

CONTROL	DESCRIPTION	NOTES
KNOB 1	Harmony / shift	Meaning follows SW2. **SW2=UP**: fixed interval, K1 = ±12 semitones. **SW2=MID**: resonance pick, K1 spans the ±36-semitone scan. **SW2=DOWN**: Bode SSB frequency shifter on the wet bus, bipolar with ±2 % deadzone — CCW = down-shift (bass), CW = up-shift, exponential taper, ±1 kHz at full deflection. In SW2 DOWN the grain buffer-read pitch is forced to unison
KNOB 2	Buffer range (bipolar)	**Noon (±6 %)** = direct-texture (grain engine bypassed, K3 = micro-stutter). Off noon either way = buffer depth 100 ms → 8 s + timescale; sign = playback direction (CW forward, CCW backward). Fully CCW = deepest buffer, played backward
KNOB 3	Character / Glitch	**Grain mode**: CCW = long, slow smear (grains stretch ~0.3 → 2 s, overlap held so the rate falls — a granular multi-tap that leans on the deep buffer), CW = short/sharp/chaotic glitch. **Direct-texture mode**: micro-stutter — CCW = clean, CW = frequent choppy repeats
KNOB 4	Texture amount	Depends on SW1 position — see below
KNOB 5	Reverb / Feedback (bipolar)	**CCW** = Clouds reverb amount (0 → 1). **Center (±5 %)** = off. **CW** = ring-buffer feedback (0 → full) into the tanh saturator — ducked and self-limiting into a controlled drone, not a runaway. Reverb tail does not feed the ring buffer
KNOB 6	Mix	0 = full dry, 1 = full wet. Equal-power curve
SWITCH 1	Texture mode	**UP** — Decimator / Wavefolder bipolar (K4 CCW = max crush, noon = clean, CW = wavefold) • **MIDDLE** — Event-driven digital glitch (bipolar K4: noon = clean ±5 %, CCW = random bit-flip events, CW = random timing events — freeze / stutter / reverse; sparse near noon → continuous at the extremes via event chaining) • **DOWN** — Ringmod (K4 0 – 30 % = tremolo 1 – 15 Hz, 30 – 100 % = bell partials, pitch-tracked with keytracked LPF)
SWITCH 2	Harmony / shift	**UP** — Fixed interval (K1 = ±12 semitones above tracked note) • **MIDDLE** — Resonance (grains lock onto nearby harmonics; K1 spans ±36-semi scan) • **DOWN** — Bode SSB frequency shifter on the wet bus (inside the feedback loop). Grain buffer-read pitch forced to unison; K1 = ±1 kHz exponential

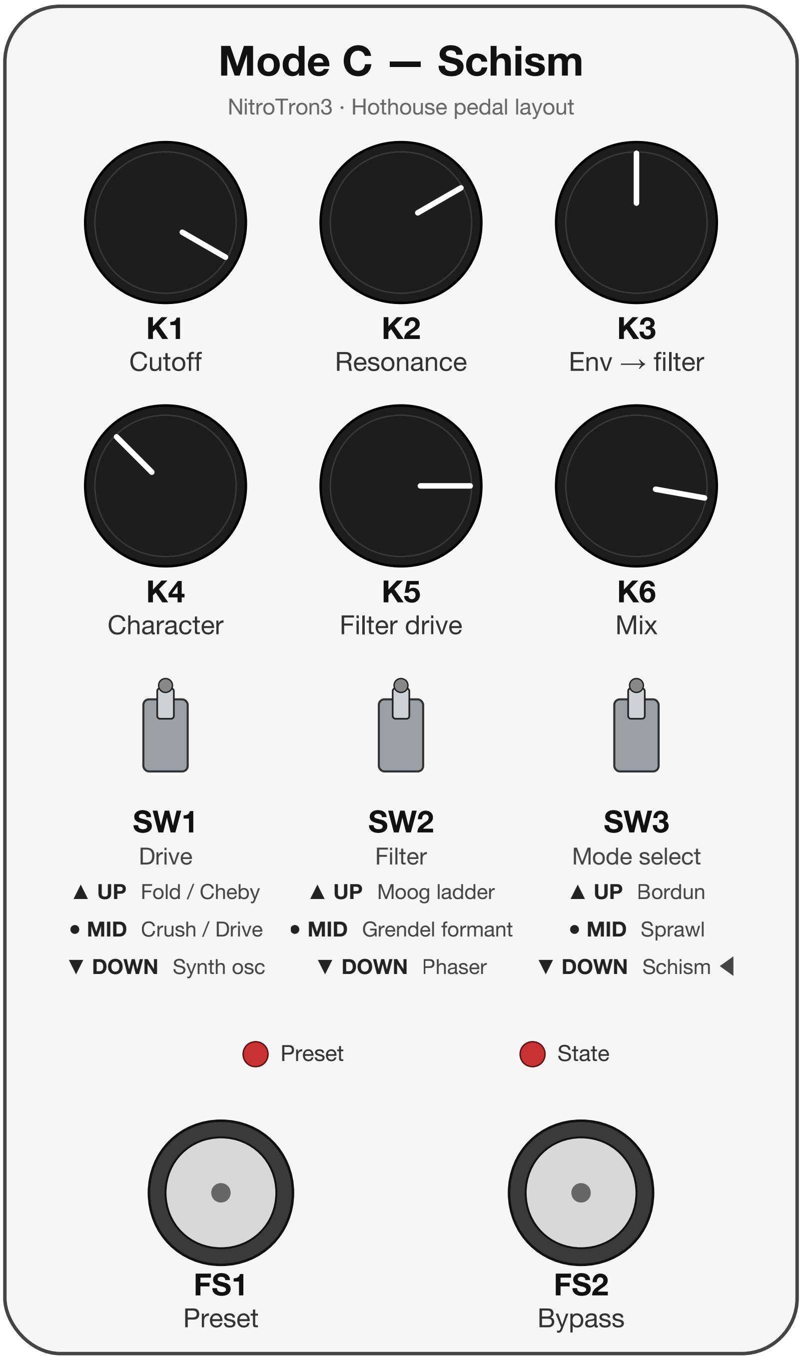
SCHISM (Mode C)

Two-stage chain: drive (SW1) → filter (SW2). K1–K3 drive the filter selected by SW2; K4 drives the source flavor selected by SW1 (wavefolder / waveshaper, bit-flipper / overdrive, or pitch-tracked synth oscillator). For SW1=UP and SW1=MID, K4 is bipolar around noon: noon = clean dry, one flavor each side. K5 is a bipolar pre-filter drive (attenuate / unity / boost, universal across all SW2 filter modes); K6 is the dry/wet mix.

Three filter flavors: a tuned **Moog ladder** (single input saturator, cutoff-tracked resonance, asymmetric drive), a vowel-pathed **Grendel formant** filter, and a 6-stage **phaser** with internal LFO. The phaser runs slightly detuned per stage (organic, less "digital") with a soft-saturated feedback loop, so K2 sweeps from a clean sweep up into a resonant bloom / controlled self-oscillation. K3 is a bipolar envelope-to-filter modulator with a center deadzone (or, on the phaser, a bipolar LFO rate + shape selector).

Three drive flavors. For **SW1=UP** and **SW1=MIDDLE**, K4 is bipolar around noon (noon = clean dry): turning it clockwise from noon brings up one flavor, counter-clockwise the other. **SW1=UP** is the sine wavefolder (K4 CW) and a Chebyshev waveshaper (K4 CCW) — an octave-up / metallic harmonic generator with a pre-shaper low-pass so it makes a clean octave instead of intermod mush. **SW1=MIDDLE** is a gated bit-flipper (K4 CW — XOR a chosen bit, swept by K4, input-envelope gated, per-bit loudness compensation) and a **Tube Screamer → tube-amp** overdrive (K4 CCW): a pre-clip high-pass into a pedal saturation stage, a touch of low-passed clean for body, then an amp saturation stage. K4-CCW is a staged master gain — the pedal/TS drive builds first, the amp drive comes in over the top of the travel; clean is via the K6 mix. **SW1=DOWN** is a pitch-tracked synth oscillator: the bass note is tracked (YIN, semitone-quantized) and an oscillator engine replaces the dry path, amplitude-gated by the env follower before it hits the filter. Here K4 is full-range (no noon split) and morphs the timbre — saw on the left half (max hypersaw at full CCW, single saw just below noon), rect on the right half (single rect just past noon, pulse-width modulated at full CW).

The wet path runs through a 2-band post-filter peak limiter — the low end is preserved so bass fundamentals don't duck under resonance peaks.



Controls

CONTROL	DESCRIPTION	NOTES
KNOB 1	Filter "where"	SW2=UP: Moog cutoff (20 Hz – 8 kHz, exponential). SW2=MID: Grendel vowel path (CCW = oo dark/ closed, CW = ee bright/open). SW2=DOWN: phaser notch centre

CONTROL	DESCRIPTION	NOTES
KNOB 2	Filter "how much"	SW2=UP: Moog resonance (0 → self-osc, sqrt curve so the lower half is audible). SW2=MID: Grendel size (mouth scale, ×0.5 → ×1.6). SW2=DOWN: phaser feedback (clean sweep → resonant bloom → controlled self-oscillation at full CW)
KNOB 3	Env / LFO modulation	Runs through a response curve (fine near noon, coarse toward the extremes). SW2=UP: bipolar env-to-cutoff (passive-bass scaled). SW2=MID: bipolar env on vowel path and size. SW2=DOWN: bipolar phaser LFO rate (sign selects shape — CCW triangle, CW sample-and-hold; magnitude = rate; centre = LFO off, static notch at K1). All with ±5 % centre deadzone
KNOB 4	Drive character	Bipolar around noon (noon = clean dry) for SW1=UP and SW1=MID. SW1=UP: CW = sine wavefold (0 → max, internal loudness comp), CCW = Chebyshev waveshaper (octave-up / metallic). SW1=MID: CW = gated bit-flipper (XOR bit position, env-gated), CCW = Tube-Screamer→tube-amp overdrive (staged master gain — pedal/TS drive builds first, amp drive enters over the top of the travel; clean via K6). SW1=DOWN: synth-osc timbre (full-range) — CCW half = saw (max hypersaw at fully CCW → single saw plateau just below noon), CW half = rect (single rect just past noon → max PWM at full CW; depth ramps in fast, then LFO rate)
KNOB 5	Filter drive (bipolar)	**CCW** attenuates (~−12 dB at full CCW). **Noon** is unity. **CW** boosts up to 8× hot. Sets the Moog ladder's input drive; pre-tanh in front of Grendel and the phaser. Moog and Grendel have a fixed internal pad so noon sits in their clean sweet zone. On the Moog ladder, K5 from just before noon up to full CW also fades in audio-rate cutoff self-FM (modulated by the filter input) for a gritty, vocal resonance
KNOB 6	Mix	0 = full dry, 1 = full wet. Equal-power curve
SWITCH 1	Drive	**UP** — Sine wavefolder (K4 CW) / Chebyshev waveshaper (K4 CCW), noon = clean • **MIDDLE** — Gated bit-flipper (K4 CW, env-gated) / Tube-Screamer→tube-amp overdrive (K4 CCW), noon = clean • **DOWN** — Pitch-tracked synth oscillator (K4 = saw ⍰ rect timbre morph)
SWITCH 2	Filter	**UP** — Moog ladder (K1 cutoff, K2 resonance, K3 env) • **MIDDLE** — Grendel formant (K1 vowel path, K2 size, K3 env on path) • **DOWN** — Phaser (K1 notch centre, K2 feedback, K3 LFO rate/shape)

Footswitches and Presets

The footswitches and the preset system work the same way in every mode.

Footswitches

CONTROL	DESCRIPTION
FOOTSWITCH 1	**Short press**: cycle Manual → 1 → … → 8 → Manual (or reload the current preset if dirty). **Long press (700 ms)**: jump to Manual
FOOTSWITCH 2	**Short press**: toggle bypass. **Long press (700 ms)**: enter save mode, or confirm save if already in save mode. **Short press in save mode**: cancel save
FS1 + FS2 short tap	Cycle the active bank (1 → 2 → 3 → 1). Both LEDs play a Roman-numeral burst confirming the new bank. Also works inside save mode to retarget the save into another bank
FS1 + FS2 held 2 s	Enter DFU bootloader for flashing new firmware (both LEDs alternate for 1.2 s before reset)

Indicator LEDs

LED	DESCRIPTION
LED 1 (left)	**Preset indicator.** Off = Manual mode. Otherwise a Roman-numeral blink pattern shows the preset number (I = short, V = long: I, II, III, IV, V, VI, VII, VIII). In save mode, shows the target slot
LED 2 (right)	**State indicator.** Solid = active, off = bypassed, rapid flash = dirty (preset edited but not saved), fast blink = save mode armed, burst = save confirmed
Both LEDs	**Bank-switch burst.** On bank change, both LEDs flash a Roman-numeral pattern of the new bank number (I / II / III, each pulse filled with deterministic fast flicker so it's visually distinct from a preset blink) for ~1.6 s, then return to their normal display

Preset behavior

- **One global edit buffer**, shared across all modes. Manual mode is fully WYSIWYG — what the panel shows is what plays.
- **3 banks × 8 slots = 24 reachable presets.** Each slot stores its own mode, knobs, SW1 and SW2, so cycling presets can swap mode mid-set.

- **Bank cycling** (FS1+FS2 short tap, 1 → 2 → 3 → 1): in manual the bank changes silently; on a saved preset the same slot number loads from the new bank; in save mode the save target shifts into the new bank (cross-bank save).
- **Dirty marking.** Adjusting any knob or switch — including SW3 — while on a saved preset dirties it (LED 2 rapid flash). Short-press FS1 while dirty to reload the preset and discard edits.
- **Save flow.** Long-press FS2 to enter save mode (LED 2 fast blink; LED 1 shows the target slot). Short-press FS1 to cycle the target slot; FS1+FS2 short tap to cycle the bank; long-press FS2 to confirm (LED 2 burst, returns to normal mode with the preset now clean); short-press FS2 to cancel.
- **Power-cycle behavior.** The pedal restores the full state on boot: active bank, active preset, the edit buffer (including mode), dirty flag. State is written to flash on a debounced 2-second timer after the last change — no writes happen when nothing is changing.
- **Migration.** Presets saved under older firmware (one bank per mode) are migrated on first boot of the new firmware: Mode A's slots → Bank 1, Mode B's slots → Bank 2, Mode C's slots → Bank 3, with each slot tagged with its source mode. No data loss.